

Mon

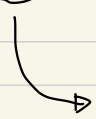
ch 7. 微分方程

$$\begin{aligned} x+1 &= 2 \\ ax &= b \\ ax^2+bx+c &= 0 \end{aligned}$$

$$\begin{aligned} x &= 1 \\ x^2 &= 1 \Rightarrow x = \pm 1 \end{aligned}$$

微分积分方程:

积分方程?



$$y = f(x) = \sin x$$

$$dy = f(x) dx = \cos x dx$$

$$\frac{d^2y}{dx^2} = (\sin x)'' = -\sin x = -y$$

(若假设 f(x) 未知)

$$y'' + y = 0$$

方程. 导数方程?

$$\frac{d^2y}{dx^2} + y = 0$$

例: $y'(x) = g(x), \quad y(x) = \int g(x) dx + C$

是 ODE 的微分方程, 常微分方程 ODE

下册: 偏微分方程 PDE

例: $y'(x) + \int_0^x y(t) dt = 0 \Leftrightarrow y''(x) + y(x) = 0$
微分积分方程 $y = \text{阿丁哥}$

Tip: 从已知列求和

问题: 方程的解. $\begin{cases} \text{① 存在吗?} \\ \text{② 唯一吗?} \end{cases} \begin{cases} \text{Yes. 怎么求} \\ \text{No. 怎么构造.} \end{cases}$

$$\begin{aligned} y'' + y &= 0 \\ y(x) &= \underline{C_1} \sin x + \underline{C_2} \cos x \\ &\uparrow \\ &\text{是解. 且是所有解} \\ &\text{(通解)} \end{aligned}$$

§ 7.1 ODE 基本概念

① 阶 : ODE 中含 $y^{(n)}$ 的最高阶为 n .

n 阶 ODE 一般形式 $F(x, y, y', \dots, y^{(n)}) = 0$, (隐式)

$y^{(n)} = G(x, y, y', \dots, y^{(n-1)})$, (显式)

eg. $y'' + y = 0$, $y'' = -y$, $\sin(y'') + \cos(y) = 0$

② 解, $y = f(x)$ 满足 ODE, 则称 $y = f(x)$ 是 ODE 的一个解.

③ 通解. 包含所有的解.
解的集合 $\rightarrow$

④ 适定性
初值条件. $y(0) = y_0, y'(0) = y'_0, \dots, y^{(n-1)}(0) = y^{(n-1)}_0$
 n 个条件

边值条件. eg. $y(0) = y_0, y(1) = y_1, \dots$

混合条件, 初边值条件

(存在性)

$\rightarrow$ 适定性. 稳定性, 正则性...

$$(y'(x))^2 + (y''(x))^2 = 0$$

$$\Downarrow$$

$$y(x) \equiv 0$$

$$F = ma, F = mg$$

$$m y'' = mg, \quad y'' = g$$

①. $t = t_0, y(t_0)$ 边值条件

②. $t = t_1, y'(t_0), y(t_1)$

eg. $y'' = g, y(0) = y_0, y'(0) = v_0$

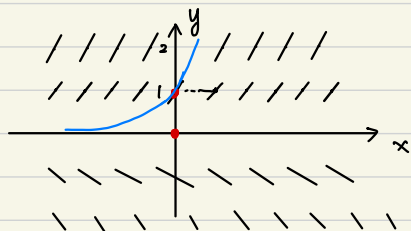
积分两次

初值条件
 $y(t) = \frac{1}{2}gt^2 + C_1t + C_2$

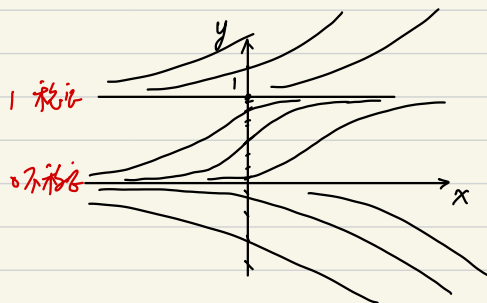
⑤ 积分曲线

↳ 解的分布图景

$$y'(x) = y(x), \quad y(x) = c e^x$$



$$y'(x) = y(x)(1 - y(x))$$



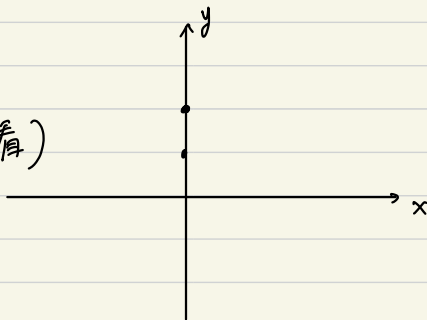
不稳定解
也可画积分曲线
(还画地面)

补充: n 阶 ODE, 要给出 n 个条件. (一般来说)

↳ 通解含有 n 个可任意常数. $y = \Phi(x; c_1, c_2, \dots, c_n)$

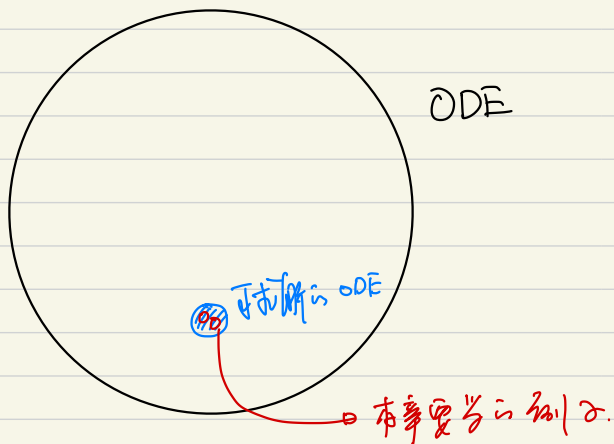
$$F(x, y, y', \dots, y^{(n)}) = 0$$

(待学了方程组之后再回过头来看)



精确
可求, 求解的 ODE 很少!

精确解.



补充

Stolz 定理.

$$\lim \frac{x_n}{y_n} = \lim \frac{\overset{=\Delta x_n}{\underbrace{x_n - x_{n-1}}}}{\underset{=\Delta y_n}{\underbrace{y_n - y_{n-1}}}} \quad (\text{有条件})$$

Tip: 对比洛必达法则. "0" : " $\frac{0}{0}$ " : " $\frac{\infty}{\infty}$ " 不定式

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)} \quad (\text{有条件})$$

和原上一样.

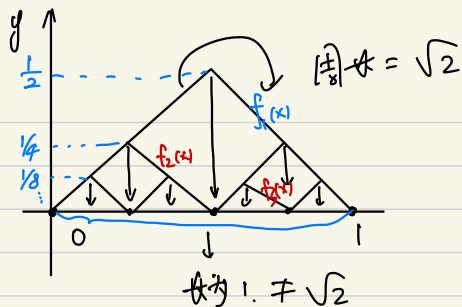
"0" : $\left. \begin{array}{l} x_n \rightarrow 0 \\ y_n \rightarrow 0, y_n \text{ 单调.} \\ \lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} \text{ 存在, 记为 } A \end{array} \right\} \Rightarrow \lim_{n \rightarrow \infty} \frac{x_n}{y_n} \text{ 存在且等于 } A$

" $\frac{\infty}{\infty}$ " : $\left. \begin{array}{l} y_n \nearrow +\infty \\ \lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} \text{ 存在, 记为 } A \end{array} \right\} \Rightarrow \lim_{n \rightarrow \infty} \frac{x_n}{y_n} = A$

例: $\lim_{n \rightarrow \infty} \frac{1^p + \dots + n^p}{n^{p+1}} = \lim_{n \rightarrow \infty} \frac{n^p}{n^{p+1} - (n-1)^{p+1}} = \frac{1}{p+1}$

$$\int_0^1 x^p dx = \frac{1}{p+1}$$

补充



$f_1(x), f_2(x), \dots$ 都连续, $f_n'(x) = \pm 1$, if $2^n x \notin \mathbb{Z}$

$$\lim_{n \rightarrow \infty} f_n(x) = 0, \quad \forall x \in [0, 1]$$

$$(f_n(x) \leq \frac{1}{2^n})$$

$$\lim_{N \rightarrow \infty} \sum_{n=1}^N q^n \underbrace{2^n f_n(x)}_{\leq 1} = G(x), \quad \text{收敛于 } G$$

$0 < q < 1$.

$\frac{1}{2} < q < 1$ 时, $G(x)$ 处处不可导. (但处处连续)

eg. $q = \frac{3}{4}$

理解: 若可导的话, 可考察 $\lim_{N \rightarrow \infty} \sum_{n=1}^N q^n 2^n f_n'(x)$

$$= \lim_{N \rightarrow \infty} \sum_{n=1}^N (2q)^n \underbrace{f_n'(x)}_{=\pm 1}, \quad \text{但这是不收敛的}$$

eg. $\lim_{N \rightarrow \infty} \sum_{n=1}^N \left(\frac{3}{2}\right)^n a_n$ 不论 a_n 取什么样的正负号
都不可收敛.

$$1 + q + \dots + q^N = \frac{q^{N+1} - 1}{q - 1}$$

$$\text{由于 } \left| \sum_{n=0}^N \left(\frac{3}{2}\right)^n a_n \right| \geq \left(\frac{3}{2}\right)^N - \frac{\left(\frac{3}{2}\right)^{N+1} - 1}{\frac{3}{2} - 1} \geq \frac{1}{2} \left(\frac{3}{2}\right)^N \rightarrow +\infty$$

§ 7.2 可分离变量 的 ODE

$$f(x, y) = \sum_{i=1}^{+\infty} g_i(x) h_i(y)$$

求解过程:

$$\frac{dy}{dx} = g(x)h(y), \quad \int \frac{dy}{h(y)} = \int g(x) dx, \quad \boxed{h(y) \neq 0} \quad \begin{matrix} 1^\circ \\ 2^\circ \end{matrix} \quad \begin{matrix} h(y) = 0 \\ y = c \end{matrix}$$
$$\int \frac{dy}{h(y)} = K(y) + C, \quad \int g(x) dx = G(x) + C \quad \Rightarrow \quad K(y) = G(x) + C$$

隐式通解.

例: $y' = \lambda y$.

$$\begin{aligned} y' &= \lambda y \\ \frac{dy}{dx} = \lambda y &\begin{cases} y=0 & \cdot \quad \frac{dy}{dx}=0, \quad y=0 \text{ 恒成立} \\ y \neq 0 & \cdot \quad \frac{dy}{y} = \lambda dx, \quad \int \frac{dy}{y} = \lambda \int dx \Rightarrow |y| = e^{\lambda x + C_1} = |C|e^{\lambda x} \\ & \parallel \quad \parallel \\ & |y| \quad \lambda x + C_1 \end{cases} \\ &\Rightarrow y = Ce^{\lambda x}, \quad C \neq 0. \end{aligned}$$

故通解为 $y = Ce^{\lambda x}, C \in \mathbb{R}.$

12a). ① $\frac{dy}{dx} = \frac{y}{x}$, ② $\frac{dy}{dx} = \frac{x}{y}$, ③ $\frac{dy}{dx} = xy$, ④ $\frac{dy}{dx} = x$, ⑤ $\frac{dy}{dx} = y$

$\Downarrow$ $\Downarrow$

$y = \frac{1}{2}x^2 + C$ $y = Ce^x$

$$\textcircled{1} \frac{dy}{dx} = \frac{y}{x} \Rightarrow \int \frac{dy}{y} = \int \frac{dx}{x} \Rightarrow |y| = |x|e^c \Rightarrow y = kx, k \in \mathbb{R} \\ \text{||} \quad \text{||} \\ \ln|y| \quad \ln|x| + c \\ (k=0 \text{ 时也是解}).$$

$$\textcircled{2} \frac{dy}{dx} = \frac{x}{y} \Rightarrow \int y dy = \int x dx \Rightarrow y^2 = x^2 + c \\ \text{||} \quad \text{||} \\ \frac{1}{2}y^2 \quad \frac{1}{2}x^2 + c_1$$

注: 草稿稿 vs 写作业

$$\textcircled{3} \frac{dy}{dx} = xy \Rightarrow \int \frac{dy}{y} = \int x dx \Rightarrow y = c e^{\frac{1}{2}x^2}, c \in \mathbb{R} \\ \text{||} \quad \text{||} \\ \ln|y| \quad \frac{1}{2}x^2 + c_1 \\ (c=0 \text{ 时也是解})$$



总结: 可分离变量的微分方程. 求解很简单. 主要是不适和分离计算

★ 若 ODE 不是可分离变量的. 怎么办?

↳ 可否 转化 为可分离变量的?

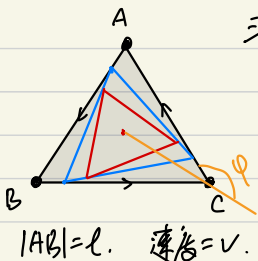
“换元” 或 做代换

齐次方程. ✓

线性方程. ✓

其他特殊情形.

例



三小物. 轨迹? 对称螺旋线

推导① t 时刻: $x_A(t), y_A(t), x_B(t), y_B(t), x_C(t), y_C(t)$.

$$\vec{v}_A(t) = \begin{pmatrix} \dot{x}_A(t) \\ \dot{y}_A(t) \end{pmatrix} \parallel \begin{pmatrix} x_B(t) - x_A(t) \\ y_B(t) - y_A(t) \end{pmatrix}, \quad \vec{v}_B(t) = \dots, \quad \vec{v}_C(t) = \dots$$

推导②. 用极坐标 $\rho(\theta), \rho'(\theta)$

$$\frac{\rho(\theta)}{\rho'(\theta)} = \tan \varphi = \tan\left(\frac{5\pi}{6}\right) = -\frac{1}{\sqrt{3}}$$

φ 始终为 $150^\circ = \frac{5\pi}{6}$

$$\rho'(\theta) = -\sqrt{3} \rho(\theta) \Rightarrow \rho(\theta) = C e^{-\sqrt{3}\theta}, \quad C \in \mathbb{R}$$

两小物一直在跑.

加上初值 $\rho(0) = l/\sqrt{3}$ 得 $\rho(\theta) = \frac{l}{\sqrt{3}} e^{-\sqrt{3}\theta}$

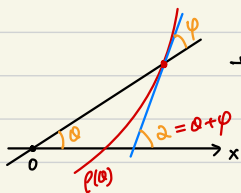
理解: 计算弧长.

有瑕

$$L = \int_0^{+\infty} \sqrt{\rho(\theta)^2 + \rho'(\theta)^2} d\theta = C \sqrt{1+a^2} \int_0^{+\infty} e^{-a\theta} d\theta = C \frac{\sqrt{1+a^2}}{a} = \frac{2l}{3}$$

$$MCP \text{ 相遇时间 } T = \frac{L}{v} = \frac{2l}{3v} \quad (\text{也可用初等办法求得})$$

补充



$$\tan \varphi = \frac{\rho(\theta)}{\rho'(\theta)}$$

$$\begin{cases} x'(\theta) = \rho'(\theta) \cos \theta - \rho(\theta) \sin \theta \\ y'(\theta) = \rho'(\theta) \sin \theta + \rho(\theta) \cos \theta \end{cases}$$

$$\frac{dy}{dx} = \tan \alpha$$

$$= \frac{\rho' \sin \theta + \rho \cos \theta}{\rho' \cos \theta - \rho \sin \theta}$$

$$= \frac{\tan \theta + \frac{\rho}{\rho'}}{1 - \tan \theta \frac{\rho}{\rho'}}$$

$$= \tan(\theta + \varphi)$$

$$\tan(\theta + \varphi)$$

$$\frac{\tan \theta + \tan \varphi}{1 - \tan \theta \tan \varphi}$$

$$\tan \varphi = \frac{\rho}{\rho'}$$

§7.3 齐次方程

eg. x, y, x^2, y^2, xy

也可写成 $\frac{dx}{dy} = 4(\frac{x}{y})$

$$P(x,y)dx + Q(x,y)dy = 0 \Rightarrow \text{齐次}$$

$$\frac{dy}{dx} = \varphi\left(\frac{y}{x}\right)$$

eg. $\frac{dy}{dx} = \frac{xy - y^2}{x^2 - 2xy}$

$$\varphi(z) = \frac{z - z^2}{1 - 2z}$$



求解: 引入 $u = \frac{y}{x}$ (变量替换)

$y = ux$ 求导分/求导(x)

$$\frac{dy}{dx} = u + x \frac{du}{dx} = \varphi(u)$$

$$\int \frac{du}{\varphi(u)-u} = \int \frac{dx}{x}$$

H(u) ln|x| + C

$$x \frac{du}{dx} = \varphi(u) - u \quad (\text{分离变量})$$

(注: $\varphi(u)-u=0$ 时, $u=c$, 即 $y=cx$)

通解: $H(\frac{y}{x}) = \ln|x| + C$

例 | $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{y^2}{xy - x^2} = \frac{(y/x)^2}{y/x - 1}$

$$\varphi(u) = \frac{u^2}{u-1}$$

$$\int \frac{du}{\varphi(u)-u} = \ln|x| + C,$$

$$\int (1 - \frac{1}{u}) du = u - \ln|u|$$

$$\frac{1}{\varphi(u)-u} = \frac{1}{\frac{u^2}{u-1}-u} = \frac{u-1}{u}$$

$$u - \ln|u| = \ln|x| + C,$$

$$u = y/x$$

$$e^u / |u| = k|x| \quad \text{即} \quad e^u = k u x \Rightarrow \underline{e^{y/x} = k y}$$

$$\text{通解为 } y = c e^{y/x} \quad (\text{隐函数}), \quad c \in \mathbb{R}$$

($c=0$ 时, 也是解)

(★作业和考试时, 要加文字描述)

②: 另或 $\frac{dy}{dx} = \varphi(\frac{y}{x})$ 或 $\frac{dx}{dy} = \psi(\frac{x}{y})$. 哪一个更容易计算? 试一试.

补充

$$\frac{dy}{dx} = f\left(\frac{ax+by+c}{a_1x+b_1y+c_1}\right), \text{ 可化为齐次的. (page 312)}$$

eg: $\frac{d(y+k_1)}{d(x+k_2)} = \frac{dy}{dx} = -\frac{2x+y-4}{x+y-1}$ 把 $x-3, y+2$

$$\frac{d(y+2)}{d(x-3)} = -\frac{2(x-3) + (y+2)}{(x-3) + (y+2)}$$

技巧: $X = x + k_2, Y = y + k_1$, 变为 X, Y 的齐次方程

取法: 解一个代数方程 (常数项变为0)

两次代换可以合到一块

$$X = x + k_2$$

$$u = \frac{y+k_1}{x+k_2} = \frac{y+k_1}{x}$$

方程变为

$$\frac{du}{dX} = F(X, u)$$

注: 知道求解思路即可.

(具体求解, 可用 Mathematica 等软件)

② 若 $\frac{a}{a_1} = \frac{b}{b_1}$, 则 $f\left(\frac{ax+by+c}{a_1x+b_1y+c_1}\right) = \tilde{f}(ax+by+c)$

可取 $v = ax+by+c$ 得 $\frac{dv}{dx} = a + b \frac{dy}{dx} = a + b \tilde{f}(v)$ 为分离变量情形.

§7.4 一阶线性 ODE

$$y = kx + b$$

$$y'(x) = -p(x)y + Q(x) \quad \text{关于未知函数 } y(x) \text{ 是线性的.}$$

证: $y_1(x), y_2(x)$ 都是解. 则 $k_1 y_1(x) + k_2 y_2(x)$ 也是解

求解: 1° 简单情形: $Q(x) = 0$, 方程变为 $y'(x) = -p(x)y$

$$\text{解为 } y(x) = C \exp(-\int p(x) dx), \quad C \in \mathbb{R}$$

2° $Q(x) \neq 0$. 变量替换 $C(x) = \frac{y(x)}{\exp(-\int p(x) dx)} = y(x) \exp(\int p(x) dx)$
 常数变易
 每 $C(x)$ 满足 u_3 方程.

$$y(x) = C(x) \exp(-\int p(x) dx), \quad \text{求导}$$

$$\begin{aligned} y'(x) &= C'(x) \exp(-\int p(x) dx) + \underbrace{C(x) \exp(-\int p(x) dx)}_{y(x)} \cdot (-p(x)) \\ &= -p(x)y + Q(x) \end{aligned}$$

$$C'(x) = Q(x) \exp(\int p(x) dx) \Rightarrow C(x) = \int Q(x) \exp(\int p(x) dx) dx + \tilde{c}$$

$$\text{代回求得 } y(x) = \left(\int Q(x) \exp(\int p(x) dx) dx + \tilde{c} \right) \exp(-\int p(x) dx)$$

$$\begin{aligned} \frac{dy}{dx} &= -p(x)y \\ \int \frac{dy}{y} &= \int -p(x) dx \\ \ln|y| &= -\int p(x) dx + C_1 \\ y &= C \exp(-\int p(x) dx) \\ &= C e^{-\int p(x) dx} \end{aligned}$$

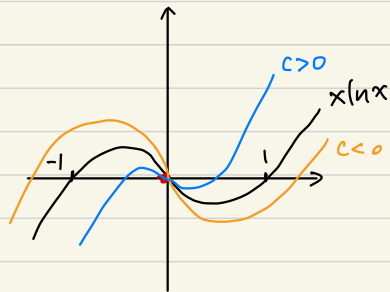
Tip: $e^x = \exp(x)$

例: $\frac{dy}{dx} = \frac{y}{x} + 1$ 注: $\frac{y}{x} + 1 = \frac{x+y}{x}$ 齐次方程

$= \frac{1}{x} \cdot y + 1$, $p(x) = -\frac{1}{x}$, $Q(x) = 1$. 线性方程

$$y(x) = \left(\int \underbrace{1}_{\frac{1}{R(x)}} \exp\left(\int \underbrace{-\frac{1}{x}}_{\frac{1}{R(x)}} dx + C_0\right) dx + C_1 \right) \exp\left(\int \underbrace{\frac{1}{x}}_{\frac{1}{R(x)}} dx\right)$$

$$= \left(\int \frac{1}{x} \frac{dx}{1} + C_0 \right) R(x) = x(\ln|x| + C)$$



补充: 例: 贝努利方程 $\frac{dy}{dx} = a(x)y + f(x)y^\alpha$, $\alpha \in \mathbb{R}$, $\alpha = 0$ 线性, $\alpha = 1$ 分离变量

做替换: $z = y^{1-\alpha}$, $\left(\frac{dy}{y^\alpha}\right) = (a(x)y^{1-\alpha} + f(x))dx$ 其他

↓
变为线性方程 $\frac{dz}{dx} = (a(x)z + f(x))(1-\alpha)$, 使用线性方程解公式.

例: $y' = y + y^2$, $z = \frac{1}{y}$, $z' = -z - 1$, $\frac{dz}{z+1} = -dx$, $\ln|z+1| = -x+C$
 $z = ke^{-x} - 1$

$y = \frac{1}{z} = \frac{1}{ke^{-x} - 1}$, $k \in \mathbb{R}$.

补充

一个特殊性的 ODE.

$$\frac{dy}{dx} = x - y^2$$

, 没法求解

Liouville
刘维氏

$\int e^{-x^2} dx$ 不是初等函数

(一) Riccati 方程

$$\frac{dy}{dx} = -y^2 + c x^m$$

$m=1$

$m = 0, -2, \frac{-4k}{2k \pm 1}, k=1, 2, \dots$ 时, 解可以求出来

$m = \text{其他}$ 求不出来

补充

- 711 ODE.

$F(x, y, \frac{dy}{dx}) = 0$. 隐式 $\rightarrow$ 特殊情形可求解.

$\frac{dy}{dx} = f(x, y)$ 显式

① $y = f(x, y')$

② $x = f(y, y')$

③ $F(x, y') = 0$

④ $F(y, y') = 0$

⑤ $F(y') = 0$

方法: 取 $p = \frac{dy}{dx}$ 为参数. 最终得 (x, y) 的参数曲线方程

例: $(\frac{dy}{dx})^3 + 2x(\frac{dy}{dx}) - y = 0$

取 $p = \frac{dy}{dx}$, 则 $y = p^3 + 2xp$, 两边求导 (关于 x)

$p = \frac{dy}{dx} = 3p^2 \frac{dp}{dx} + 2x \frac{dp}{dx} + 2p$

$\frac{dp}{dx} = \frac{-p}{3p^2 + 2x}$

$(3p^2 + 2x) dp + p dx = 0$ ($p(x)$ 满足此方程)

$\frac{dx}{dp} = \frac{3p^2 + 2x}{-p} = -3p - \frac{2}{p}x$
线性方程

可求解得 $\frac{3}{4}p^4 + xp^2 = C$, $x = \frac{C - \frac{3}{4}p^4}{p^2}$

合并 $\begin{cases} x = \frac{C - \frac{3}{4}p^4}{p^2} \\ y = p^3 + 2xp = p^3 + \frac{2C - \frac{3}{2}p^4}{p} \end{cases}$

为参数曲线

(解的一种形式)

参数方程
形式的解

补充

例: $y(1+y^2) = C$ (Ref: 2022.11.02 昂泰陈成)

方法1: 若 $y \neq 0$, 则 $y^2 = \frac{C}{y} - 1$ (注 $y \equiv 0$ 是一解)

$$\frac{dy}{dx} = \pm \sqrt{\frac{C}{y} - 1} \Rightarrow \frac{dy}{\sqrt{\frac{C}{y} - 1}} = \pm dx$$
$$= \sqrt{\frac{C}{y} - 1} - C \arctan \sqrt{\frac{C}{y} - 1} = \pm x + \tilde{C}$$

方法2: 取 $p = y'$ 则 $y(1+p^2) = C$, $y = \frac{C}{1+p^2}$.

关于 x 求导得 $p = y' = \frac{d}{dx} \left(\frac{C}{1+p^2} \right) = \frac{-2Cp}{(1+p^2)^2} \frac{dp}{dx} \Rightarrow \frac{dp}{dx} = \frac{(1+p^2)^2}{-2C}$

$$\frac{dx}{dp} = \frac{-2C}{(1+p^2)^2} \Rightarrow \begin{cases} x(p) = \int \frac{-2C}{(1+p^2)^2} dp = -C \left(\frac{p}{1+p^2} + \arctan(p) \right) + \tilde{C} \\ y(p) = \frac{C}{1+p^2} \end{cases}$$

$\in (-\frac{\pi}{2}, \frac{\pi}{2})$

(不太简洁) 换参法 $\theta = \arctan(p)$

则 $p = \tan \theta$, $\frac{p}{1+p^2} = \frac{\tan \theta}{1+\tan^2 \theta} = \sin \theta \cdot \cos \theta$

$$\begin{cases} x(\theta) = -C(\theta + \sin \theta \cdot \cos \theta) + \tilde{C} \\ y(\theta) = C \cos^2 \theta \end{cases}$$

$$\frac{1}{1+p^2} = \frac{1}{1+\tan^2 \theta} = \cos^2 \theta$$

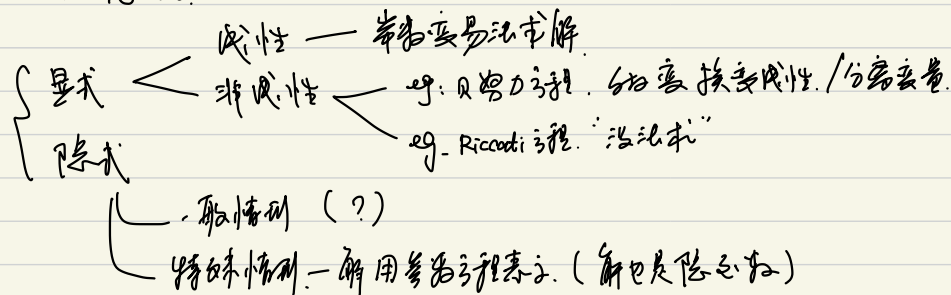
(还是不太简洁) 利用 $\sin \theta \cdot \cos \theta = \frac{1}{2} \sin(2\theta)$

$$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$$

$$\begin{cases} x(\theta) = -\frac{C}{2}(2\theta + \sin(2\theta)) + \tilde{C} \\ y(\theta) = \frac{C}{2}(1 + \cos(2\theta)) \end{cases} \xrightarrow{\text{再令}} \begin{cases} x(t) = \frac{C}{2}(t - \sin t) + \tilde{C} \\ y(t) = \frac{C}{2}(1 - \cos t) \end{cases}$$

$t = \pi + 2\theta \in (0, 2\pi)$ 提醒

一阶 ODE 总结



n 阶 ODE: $F(x, y, \dots, y^{(n)}) = 0$.

求解思路: 降阶.

补充

§ 7.5. 可降阶的高阶 ODE.

① $y^{(n)} = f(x)$, 易. $y^{(n-1)} = \int f(x) dx + C_1$, $y^{(n-2)} = \int (\int f(x) dx) dx + C_1 x + C_2$, ...

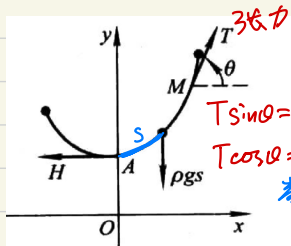
② $y'' = f(x, y')$ } 取 $y' = p$. ② $p' = y'' = f(x, p)$ 变为 $\frac{dp}{dx} = f(x, p)$ - 1 阶方程
 若求得通解 $p(x, C_1)$, 则后方程为 $y = \int p(x, C_1) dx + C_2$

③ $y'' = f(y, y')$ } ③ $\frac{dp}{dy} = \frac{dp}{dx} \frac{dx}{dy} = y'' \cdot \frac{1}{p} = \frac{f(y, p)}{p}$ 变为 $p \frac{dp}{dy} = f(y, p)$ (-1 阶方程)
 若通解为 $p(y, C_1)$, 则 $x = \int \frac{dy}{p(y, C_1)} + C_2$

$y'' = f(x, y, y')$

补充

例: 悬链线



1° 建立方程.

$$\Rightarrow \tan \theta = \frac{s}{a} \Rightarrow y' = \frac{1}{a} \int_0^x \sqrt{1+y'^2} dx \Rightarrow y'' = \frac{1}{a} \sqrt{1+y'^2}$$

(积分微分方程)

定解条件: $y(0) = y_0, y'(0) = 0$

page 324. 例4.

2° 求解 ODE. $y' = p. \quad \frac{dp}{dx} = y'' = \frac{1}{a} \sqrt{1+p^2}, \quad \frac{dp}{\sqrt{1+p^2}} = \frac{dx}{a}$

$$\Rightarrow \ln(p + \sqrt{1+p^2}) = \frac{x}{a} + C_1, \quad \text{由于 } p(0) = y'(0) = 0 \text{ 知 } C_1 = 0$$

$$\text{即可解得 } p = \frac{1}{2} (e^{\frac{x}{a}} - e^{-\frac{x}{a}})$$

$$\Rightarrow y = \frac{a}{2} (e^{\frac{x}{a}} + e^{-\frac{x}{a}}) + C_2, \quad C_2 \text{ 可由 } y(0) = y_0 \text{ 决定}$$

注: $\frac{e^x + e^{-x}}{2} = \text{ch}(x),$

$$\frac{e^x - e^{-x}}{2} = \text{sh}(x)$$

$$\text{sh}'(x) = \text{ch}(x)$$

$$y = a \text{ch}(x) + C$$

Tip: $\int \frac{dp}{\sqrt{1+p^2}} = \text{arsh}(p) + C$

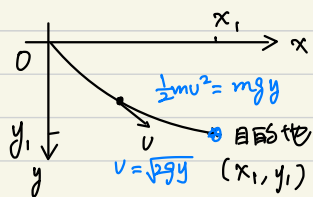
$$\text{arsh}(x) = \ln(x + \sqrt{x^2+1})$$

$$\text{arch}(x) = \ln(x + \sqrt{x^2-1})$$

悬链线 是双曲余弦函数

补充

最速降线



定解条件: $y(0) = 0$
 $y(x_1) = y_1$

1° 建立方程.

方式1: 利用费马原理 (2022.11.02), $y(1+y'^2) = C$.

方式2: 利用变分法, 先计算时间

$$T = \int_0^{x_1} \frac{ds}{v} = \int_0^{x_1} \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx$$

这是 $y(x)$ 的泛函. 记为 $T[y]$, 求极值, 要'导数=0' $\delta T = 0$

$$\lim_{\varepsilon \rightarrow 0} \frac{T[y + \varepsilon \delta y] - T[y]}{\varepsilon} = 0 \Rightarrow E-L \text{ 方程}$$

具体求导 $2yy'' + y'^2 + 1 = 0$ 是 p 方程

2° 降阶. 取 $p = y'$, 则 $y'' = p' = -\frac{1+p^2}{2y}$, 由 $p' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}$

$$\text{可得 } p \frac{dp}{dy} = -\frac{1+p^2}{2y} \Rightarrow \frac{2p dp}{1+p^2} = -\frac{dy}{y} \Rightarrow \frac{d(1+p^2)}{1+p^2} = -\frac{dy}{y} \Rightarrow \ln(1+p^2) = -\ln|y| + C,$$

$$\Rightarrow y(1+p^2) = C \text{ 即 } y(1+y'^2) = C \text{ 与费马原理所得方程一样.}$$

3° 求解 $y(1+y'^2) = C$. 如前.

$$\text{得 } \begin{cases} x(t) = \frac{C}{2}(t - \sin t) + \tilde{C} \\ y(t) = \frac{C}{2}(1 - \cos t) \end{cases}$$

思考: 由 $y(x_1) = y_1$ 如何求解 C 和 T ?

2022.12.12

Mon

§7.6 (高阿) 线性 ODE (I)

思考: $x_{n+1} = x_n + x_{n-1}$ 差分方程 } 有何区别与联系?
 $x''(t) = x'(t) + x(t)$ 微分方程
 $x^2 = x + 1$ 代数方程

本质: 都是线性的! (线性代数)

$$① \begin{pmatrix} x_n \\ x_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ x_n \end{pmatrix} =: A \begin{pmatrix} x_{n-1} \\ x_n \end{pmatrix}$$

$$② \begin{pmatrix} x(t) \\ x'(t) \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x(t) \\ x'(t) \end{pmatrix} =: A \begin{pmatrix} x(t) \\ x'(t) \end{pmatrix}$$

$$③ \begin{pmatrix} x \\ x^2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ x \end{pmatrix} =: A \begin{pmatrix} 1 \\ x \end{pmatrix}$$

↑
三个矩阵一样

矩阵乘法

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}$$

★ A 的本质是 $n \times n$

本征值, 本征向量
特征值, 特征向量

$$A \vec{x} = \lambda \vec{x}$$

$$① \begin{pmatrix} x_n \\ x_{n+1} \end{pmatrix} = \lambda \begin{pmatrix} x_{n-1} \\ x_n \end{pmatrix} \text{ 即 } x_n = \lambda x_{n-1} \Rightarrow x_n = c \lambda^n$$

$$② \begin{pmatrix} x(t) \\ x'(t) \end{pmatrix}' = \lambda \begin{pmatrix} x(t) \\ x'(t) \end{pmatrix} \text{ 即 } x'(t) = \lambda x(t) \Rightarrow x(t) = c e^{\lambda t}$$

$$③ \begin{pmatrix} x \\ x^2 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ x \end{pmatrix} \text{ 即 } x = \lambda$$

三省通过“线性”联系起来

线性函数

一元 $f(x) = ax + b$

二元 $f(x, y) = a_1x + a_2y + a_0$

三元 $f(x_1, x_2, x_3) = a_1x_1 + \dots + a_3x_3 + a_0$

① 线性 ODE. 定义

$\hookrightarrow F(x, y, y', \dots, y^{(n)}) = 0$, $y^{(n)} = f(x, y, y', \dots, y^{(n-1)})$
↑ 线性函数 ↑ 线性函数

eg. = 阶线性 ODE. $y'' = f(x, y, y') = a_1(x)y' + a_2(x)y + a_0(x)$ 变系数的.
 $a_1y' + a_2y + a_0$ 常系数的

④ 齐次 vs 非齐次 $\rightarrow a_0(x) \neq 0$. } 思考: 为什么区分这两种情况??
 $\hookrightarrow a_0(x) \equiv 0$.

② 解的性质/结构, 齐次方程的通解 = 集合 $Y = \{ y(x) \mid y' = a_1y' + a_2y \}$

“整体来看”比“具体看”容易.

* 若 $\left. \begin{matrix} y_1(x) \in Y \\ y_2(x) \in Y \end{matrix} \right\} \Rightarrow C_1y_1(x) + C_2y_2(x) \in Y$ “线性性质”
“叠加原理”

check: $\left. \begin{matrix} y_1' = a_1y_1' + a_2y_1 \\ y_2' = a_1y_2' + a_2y_2 \end{matrix} \right\} \Rightarrow (C_1y_1 + C_2y_2)' = a_1(\dots)' + a_2(\dots)$

⑤ n 阶齐次线性 ODE. 通解形式为:

$y(x) = \underbrace{C_1y_1(x) + \dots + C_ny_n(x)}_{n \text{ 个可变常数}}$, 而且 $y_1(x), \dots, y_n(x)$ “线性无关”

$\hookrightarrow C_1y_1(x) + \dots + C_ny_n(x) = 0$
 $\Rightarrow C_1 = C_2 = \dots = C_n = 0$

eg. = 阶方程. $y_1(x)$ 与 $y_2(x)$ 线性无关, 即指 $y_1(x)$ 与 $y_2(x)$ 不成比例.

① n 阶非齐次线性ODE 通解形式为 $\longrightarrow$ 记为集合 Z_n

$$y(x) = \underbrace{c_1 y_1(x) + \dots + c_n y_n(x)}_{\text{对应齐次方程的通解}} + \underbrace{y^*(x)}_{\text{特解}}$$

$$= (y_1^T, y_2^T, \dots, y_n^T) \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} + y^*(x)$$

$$\{y(x) \mid y' = a_1 y' + a_2 y + \underline{a_0}\}$$

$$\text{若 } y_1(x) \in Z, y_2(x) \in Z \Rightarrow y_1(x) - y_2(x) \in Z$$

check: 代入方程. 验证

补充:

③ n 阶线性ODE, 形式 - n 阶线性ODE (组) 3个组

(多个方程放一起)

$$y'' = a_1 y' + a_2 y + a_0 = \text{二阶}$$

$$\begin{cases} y' = p \\ p' = y'' = a_1 p + a_2 y + a_0 \end{cases} \quad \text{二阶方程组}$$

$$\underbrace{\begin{pmatrix} y \\ p \end{pmatrix}}_{\vec{z}}' = \underbrace{\begin{pmatrix} y' \\ p' \end{pmatrix}}_{\vec{z}'} = \underbrace{\begin{pmatrix} p \\ a_1 p + a_2 y + a_0 \end{pmatrix}}_{\vec{z}'} = \underbrace{\begin{pmatrix} 0 & 1 \\ a_2 & a_1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} y \\ p \end{pmatrix}}_{\vec{z}} + \underbrace{\begin{pmatrix} 0 \\ a_0 \end{pmatrix}}_{\vec{b}}$$

$$\vec{z}' = A\vec{z} + \vec{b}, \quad \text{二阶方程组 } z' = az + b$$

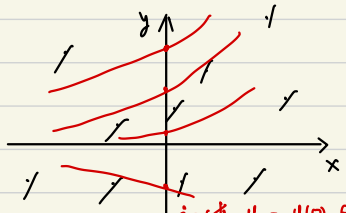
一阶方程组

可利用常变易法

可用类似方法求解.

① 理解: n 阶方程. 通解有 n 个常数

n 个一阶方程



初值 $y_0 = y(0) \in \mathbb{R}$

一个方程的情况

每个初值都对应一条积分曲线 (即一个解)

补充:

例: 验证 $y(x) = c_1 e^x + c_2 e^{2x} + \boxed{\frac{1}{12} e^{5x}}$ 满足二阶线性 ODE.

方法1: $y'(x) = c_1 e^x + 2c_2 e^{2x} + \frac{5}{12} e^{5x}$
 $y''(x) = c_1 e^x + 4c_2 e^{2x} + \frac{25}{12} e^{5x}$ } 消去 c_1 和 c_2 , 消元法.

$$y'' - 3y' + 2y = 0$$

齐次 ODE. 通解

方法2: $e^{-x} y(x) = c_1 + c_2 e^x + \frac{1}{12} e^{4x}$ } 求导

$$(e^{-x} y(x))' = \downarrow c_2 e^x + \frac{1}{3} e^{4x}$$

$$e^{-x} (e^{-x} y(x))' = c_2 + \frac{1}{3} e^{3x}$$
 } 求导

$$\boxed{(e^{-x} (e^{-x} y(x))')' = e^{3x}}$$

化简

$$\iff y'' - 3y' + 2y = e^{5x}$$

★ $y(x) = c_1 e^x + c_2 e^{2x} + \boxed{\frac{1}{12} e^{5x}}$
 是通解. 特解

$$(e^{-x} (e^{-x} y' - e^{-x} y))'$$

$$(e^{-2x} (y' - y))' = e^{-2x} [(y'' - y') - 2(y' - y)] = e^{-2x} [y'' - 3y' + 2y]$$

补充

方法3: 利用解的结构. $\tilde{y}(x) = c_1 e^x + c_2 e^{2x}$ 满足齐次 $y'' = a_1 y' + a_2 y$.

$$(D-1)y=0 \Rightarrow y'-y=0 \Leftrightarrow y'=y \quad y'=2y \Rightarrow y'-2y=0 \Rightarrow (D-2)y=0$$

$$\text{合并: } (D-1)(D-2)y=0 \Rightarrow (D^2-3D+2)y=0, \text{ 即 } y''-3y'+2y=0$$

$$y^* = \frac{1}{12} e^{5x}, \quad a_0(x) = (D^2-3D+2)y^* = \dots = e^{5x}$$

基解矩阵

补充:

$$y(x) = c_1 y_1(x) + \dots + c_n y_n(x) + y^*(x)$$

$$y^{(n)} = a_1 y^{(n-1)}(x) + \dots + a_n y^{(0)}(x) + a_0$$

→ =

$$\begin{pmatrix} y_1(x) & \dots & y_n(x) \\ y_1'(x) & \dots & y_n'(x) \\ \vdots & & \vdots \\ y_1^{(n-1)}(x) & \dots & y_n^{(n-1)}(x) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

$$\vec{z} = \begin{pmatrix} y \\ y' \\ \vdots \\ y^{(n-1)} \end{pmatrix}, \quad \vec{z}' = A \vec{z} + \vec{b}, \quad \text{通解为 } \vec{z} = \begin{pmatrix} c_1 y_1(x) + \dots + c_n y_n(x) \\ c_1 y_1'(x) + \dots + c_n y_n'(x) \\ \vdots \\ c_1 y_1^{(n-1)}(x) + \dots + c_n y_n^{(n-1)}(x) \end{pmatrix} + \vec{z}^*$$

⊕ 朗斯基行列式 (Wronski)

$$W(x) = \begin{vmatrix} y_1(x) & \dots & y_n(x) \\ \vdots & \ddots & \vdots \\ y_1^{(n-1)}(x) & \dots & y_n^{(n-1)}(x) \end{vmatrix},$$

解 $y_1, \dots, y_n$ 线性相关解 $y_1, \dots, y_n$ 线性无关

W(x) 要恒为零, 要恒不为零

↑

Tip: $W(x)$ 满足方程 $W'(x) = \text{Tr}(A(x)) W(x)$, 令 $W(x) = W(0) e^{\int \text{Tr}(A(x)) dx}$
(Liouville 定理)

注:

若 $y_1, \dots, y_n$ 不是解,

则结论不一定对.

$$y_1(x) = \begin{cases} 0, & x > 0 \\ x^2, & x \leq 0 \end{cases}$$

$$y_2(x) = \begin{cases} x^2, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$W(x) = 0$$

但 y_1, y_2 线性无关

考

★ 常系数线性 ODE 的求解 (§7.7)

例: $y'' - 3y' + 2y = 0$ 通解如何求?

草稿: $\lambda^2 - 3\lambda + 2 = 0 \Rightarrow (\lambda - 1)(\lambda - 2) = 0, \lambda_1 = 1, \lambda_2 = 2. \Rightarrow y(x) = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$

$\begin{pmatrix} y \\ y' \end{pmatrix}' = \begin{pmatrix} A \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix}, \quad \vec{z}' = A \vec{z}, \quad A \vec{z} = \lambda \vec{z}, \text{ 特征值, 特征方程}$

简单法: $y(x) = e^{\lambda x}$ 代入方程. ~~$e^{\lambda x}$~~ $(\lambda^2 - 3\lambda + 2) = 0$

得 $y_1(x) = e^{\lambda_1 x} = e^x, y_2(x) = e^{\lambda_2 x} = e^{2x}$
 $\underbrace{\hspace{10em}}_{\text{线性无关}}$

通解为 $C_1 e^x + C_2 e^{2x}$

注: $\lambda_1 = \lambda_2$ 的情况怎么办?
法1: 做变换 $y(x) e^{-\lambda x} = u(x)$
法2: "取极限"

例: $y'' - 2y' + y = 0$ 通解?

$\lambda^2 - 2\lambda + 1 = 0, (\lambda - 1)^2 = 0, \lambda_1 = \lambda_2 = 1, \quad \cancel{C_1 e^x + C_2 e^x}$

$C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} = \tilde{C}_1 \frac{e^{\lambda_1 x} - e^{\lambda_2 x}}{\lambda_1 - \lambda_2} + \tilde{C}_2 e^{\lambda_1 x}$
 $\lambda_2 \rightarrow \lambda_1 \hookrightarrow \tilde{C}_1 x e^{\lambda_1 x} + \tilde{C}_2 e^{\lambda_1 x}$

2022.12.14
Wed

例: $y'' + k^2 y = 0$ 通解 ($k \neq 0$)

特征方程 $\lambda^2 + k^2 = 0$, $\lambda = \pm ki$. 通解 $y(x) = c_1 e^{-ki x} + c_2 e^{ki x}$, $c_1, c_2 \in \mathbb{C}$.

利用 $e^{ix} = \cos(x) + i \sin(x)$ ← 希望得到实函数.
Euler 公式

$$e^{kix} = \cos(kx) + i \sin(kx)$$

$$e^{-kix} = \cos(kx) - i \sin(kx)$$

$$y(x) = \tilde{c}_1 \cos(kx) + \tilde{c}_2 \sin(kx), \quad \tilde{c}_1, \tilde{c}_2 \in \mathbb{R}$$

小结: 对于 n 阶常系数线性 ODE. 求解方法.

$$y^{(n)} + \underline{a_1} y^{(n-1)} + \dots + \underline{a_n} y^{(0)} = 0$$

① 写出特征方程 $\lambda^n + a_1 \lambda^{n-1} + \dots + a_n = 0$, ② 求解得 $\lambda_1, \lambda_2, \dots, \lambda_n$

③ 写出通解 $y(x) = c_1 y_1(x) + \dots + c_n y_n(x)$. $y_1(x), \dots, y_n(x)$ 取法如下:

1°. λ_i 是实单根. $y_i(x) = e^{\lambda_i x}$

2°. $\lambda_{i1} = \lambda_{i2} = \dots = \lambda_{ik} = \lambda_i$ 是实重根. $y_{ij}(x) = e^{\lambda_i x} x^{j-1}$ 即 $e^{\lambda_i x}, x e^{\lambda_i x}, \dots, x^{k-1} e^{\lambda_i x}$

3°. $\lambda_{i1}, \lambda_{i2}$ 是一对共轭复根. $y_{i1}(x), y_{i2}(x)$ 取法 " $e^{(2 \pm i\beta)x}$ ", $e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x)$.

4°. $\lambda_{i_1}, \lambda_{i_2}, \dots, \lambda_{i_{k-1}}, \lambda_{i_k}$ 复重根. 取 $x^{j-1} e^{(\alpha \pm i\beta)x}$, $x^{j-1} e^{\alpha x} \cos(\beta x)$, $x^{j-1} e^{\alpha x} \sin(\beta x)$, $j=1, \dots, k$

注: 微分算子. $\cdot D''$ $Dy = y'$, $D^2 y = y''$, $D^n = y^{(n)}$

$$(D^n + a_1 D^{n-1} + \dots + a_n) y = 0$$

§7.7 §7.8 常系数线性 ODE

齐次 非齐次

$$y'' + a_1 y' + a_2 y = f(x)$$

- 降阶法

先变成方程组:
$$\begin{cases} y' = p \\ p' = y'' = -a_1 p - a_2 y + f \end{cases}$$

$$\begin{aligned} \vec{z}' &= A \vec{z} + \vec{F} \\ \begin{pmatrix} y \\ p \end{pmatrix}' &= \begin{pmatrix} 0 & 1 \\ -a_2 & -a_1 \end{pmatrix} \begin{pmatrix} y \\ p \end{pmatrix} + \begin{pmatrix} 0 \\ f(x) \end{pmatrix} \end{aligned}$$

解: 用常系数法

步骤: ① 考虑 $\vec{z}' = A \vec{z}$

即 $y'' + a_1 y' + a_2 y = 0$, $y(x) = C_1 y_1(x) + C_2 y_2(x)$

$p(x) = C_1 y_1'(x) + C_2 y_2'(x)$

$$\begin{pmatrix} y \\ p \end{pmatrix} = \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$\vec{\Phi}$, 基解矩阵

通解为 $\vec{z} = \vec{\Phi}(x) \vec{C}$

② 设 $\vec{z} = \vec{\Phi}(x) \vec{C}(x)$ 推导 $\vec{C}(x)$ 满足的方程

得 $\vec{z}' = A \vec{z} + \vec{F}$ 的通解为

$$\vec{z} = \vec{\Phi}(x) \left(\vec{C}_0 + \int \underbrace{\vec{\Phi}^{-1}(x)}_{\text{逆矩阵}} \vec{F}(x) dx \right)$$

$$\vec{z}' = A \vec{z} + \vec{F}$$

$$\vec{\Phi}'(x) \vec{C}(x) + \vec{\Phi}(x) \vec{C}'(x)$$

$$\underbrace{A \vec{\Phi} \vec{C}}_{\vec{z}'} + \vec{\Phi} \vec{C}'$$

$$\Rightarrow \vec{\Phi} \vec{C}' = \vec{F}$$

$$\vec{C}' = \vec{\Phi}^{-1} \vec{F}$$

$$\vec{C} = \int \vec{\Phi}^{-1} \vec{F} dx$$

$$\therefore \vec{\Phi}(x) \vec{\Phi}^{-1}(x) = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

补充

补充

例: (page 335 例3) $y'' - \frac{x}{x-1} y' + \frac{1}{x-1} y = x-1$, $((x-1)y'' - xy' + y = (x-1)^2)$

其中齐次方程的解为 $Y(x) = C_1 x + C_2 e^x$

Step 0: $\vec{z} = \begin{pmatrix} y \\ p \end{pmatrix}$, $p = y'$, $Y'(x) = C_1 + C_2 e^x$, $\vec{F} = \begin{pmatrix} 0 \\ x-1 \end{pmatrix}$

Step 1: $\vec{z} = \Phi \vec{c} = \begin{pmatrix} C_1 x + C_2 e^x \\ C_1 + C_2 e^x \end{pmatrix} = \boxed{\begin{pmatrix} x & e^x \\ 1 & e^x \end{pmatrix}} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$
(齐次)

Step 2: 非齐次 ODE 特解: $\vec{z}^* = \begin{pmatrix} y^* \\ p^* \end{pmatrix}$

$\vec{z} = \Phi(x) \vec{c}_0 + \underbrace{\Phi(x) \int \Phi^{-1}(x) \vec{F}(x) dx}$

$\downarrow \begin{pmatrix} x & e^x \\ 1 & e^x \end{pmatrix} \int \frac{1}{(x-1)e^x} \begin{pmatrix} e^x & -e^x \\ -1 & x \end{pmatrix} \begin{pmatrix} 0 \\ x-1 \end{pmatrix} dx$
 $\downarrow \frac{1}{(x-1)e^x} \begin{pmatrix} -e^x(x-1) \\ x(x-1) \end{pmatrix} = \begin{pmatrix} -1 \\ xe^{-x} \end{pmatrix}$

$\downarrow = \begin{pmatrix} x & e^x \\ 1 & e^x \end{pmatrix} \begin{pmatrix} \int -1 dx \\ \int xe^{-x} dx \end{pmatrix} = \begin{pmatrix} x & e^x \\ 1 & e^x \end{pmatrix} \begin{pmatrix} -x + C_1 \\ -(x+1)e^{-x} + C_2 \end{pmatrix} = \begin{pmatrix} -x^2 - (x+1) \\ -x - (x+1) \end{pmatrix} = \begin{pmatrix} -x^2 - x - 1 \\ -2x - 1 \end{pmatrix}$
 $\hookrightarrow p^* = y^{*'} = -2x - 1$

即通解为 $y(x) = C_1 x + C_2 e^x - (x^2 + x + 1)$

Tip:

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$\int x e^{-x} dx = -(x+1)e^{-x} + C$

一些特殊情形的简单求法 $f(x)$ 具有特殊形式

★ 若 $f(x)$ 是指数为 λ 的幂函数 $f(x) = e^{\lambda x} P_m(x)$

Tip: 这时可用待定系数法.

↳ 若 $\lambda = \alpha + i\beta$, 则取为 $e^{\alpha x} (P_e(x) \cos(\beta x) + Q_n(x) \sin(\beta x))$ 次为可不同.

做法: 3 种的特解 $y^*(x) = e^{\lambda x} \underline{R(x)}$ 多项式

↳ 1. 确定 $R(x)$ 的次数

2. 待定系数法确定多项式系数.

例: $y'' - 2y' - 3y = 3x + 1$, 一个特解. $\lambda = 0$. $P_m(x) = 3x + 1$.

Tip: 有一个多项式特解. $y^*(x) = ax + b$ (思考: 为什么是一个多项式)

$$\text{代入方程中得: } y^{*''} - 2y^{*'} - 3y^* = -2a - 3(ax + b) = 3x + 1 \Rightarrow a = -1, b = \frac{1}{3}$$

$$\text{从而特解为 } y^*(x) = -x + \frac{1}{3}.$$

例: $y'' - 5y' + 6y = \underline{x e^{2x}}$. 通解 $e^{\lambda x} P_m(x)$, $\lambda = 2$, $P_m(x) = x$

$$1^\circ \text{ 齐次方程 } y'' - 5y' + 6y = 0, \quad \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3) = 0$$

$$\text{通解为 } y(x) = C_1 e^{2x} + C_2 e^{3x}$$

2^\circ \text{ 特解 } y^*(x) = e^{\lambda x} P_m(x), \text{ } P_m(x) \text{ 为多项式 (思考: } Q(x) \text{ 取多少阶)}

$$\text{代入方程中 } y^*(x) = e^{\lambda x} (P_m'(x) + \lambda P_m(x))$$

$$y^{*''}(x) = e^{\lambda x} (P_m''(x) + 2\lambda P_m'(x) + \lambda^2 P_m(x)), \quad \lambda = 2$$

$$\hookrightarrow y^{*''} - 5y^{*'} + 6y^* = e^{\lambda x} (\underbrace{P_m''(x) + 4P_m'(x) + 4P_m(x) - 5P_m'(x) - 10P_m(x) + 6P_m(x))}_{= x e^{2x}}) = x e^{2x}$$

$$\underline{P_m''(x) - P_m'(x) = x} \quad \text{可知 } P_m(x) \text{ 取二次多项式} \quad P_m(x) = ax^2 + bx + c$$

$$\text{代入可得 } P_m(x) = -\frac{x^2}{2} + x + c. \quad 3^\circ \text{ 通解 } y(x) = C_1 e^{2x} + C_2 e^{3x} + e^{2x} \left(-\frac{x^2}{2} + x\right)$$

例: $y'' + y = x \cos 2x$ 特解. $e^{\lambda x} p_m(x)$, $\lambda = \pm 2i$, $p_m(x) = x$.

$$e^{\lambda x} R(x) = e^{\pm 2ix} R(x) \rightarrow \cos(2x) = \frac{e^{i2x} + e^{-i2x}}{2}$$

Page 351 例 3. $y^*(x) = -\frac{1}{3}x \cos 2x + \frac{4}{9} \sin 2x$.

注: 计算过程 - 取比较麻烦. (特解形式 $y^*(x) = (ax+b) \cos(2x) + (cx+d) \sin(2x)$)

小结: page 348. page 351. 有一些结论. 多次求导取法 (简)
与 'x' 是不是特征方程的根有关

“另类求解” $x \cos(2x) = \frac{d(\sin(\lambda x))}{d\lambda} \Big|_{\lambda=2}$, 考虑方程 $y'' + y = \sin(\lambda x) = \frac{e^{i\lambda x} - e^{-i\lambda x}}{2i}$

取 $y_1'' + y_1 = e^{i\lambda x}$, $y_2'' + y_2 = e^{-i\lambda x}$
 $y_1 = a e^{i\lambda x}$, $y_2 = \bar{y}_1$

$((i\lambda)^2 + 1)a = 1$
 $\Downarrow$
 $a = \frac{1}{1 - \lambda^2}$

$y_1 = \frac{1}{1 - \lambda^2} e^{i\lambda x}$

利用叠加原理 $y = \frac{1}{2i} \frac{1}{1 - \lambda^2} (e^{i\lambda x} - e^{-i\lambda x})$
 $= \frac{\sin(\lambda x)}{1 - \lambda^2}$

代回 $y^* = \frac{dy}{d\lambda} \Big|_{\lambda=2} = \frac{d}{d\lambda} \left(\frac{\sin(\lambda x)}{1 - \lambda^2} \right) \Big|_{\lambda=2}$
 $= \frac{\cos(\lambda x) \times (1 - \lambda^2) + 2\lambda \sin(\lambda x)}{(1 - \lambda^2)^2} \Big|_{\lambda=2}$
 $= \frac{-3x \cos(2x) + 4 \sin(2x)}{9}$

2022.12.16

Fri

例: 强迫振动 ODE

$$\downarrow \uparrow$$

$$\frac{d^2 x}{dt^2} + 2n \frac{dx}{dt} + k^2 x = h \sin(pt)$$

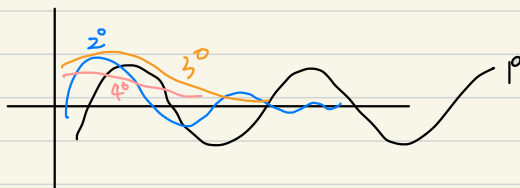
①. 齐次: $h=0$. $x'' + 2n x' + k^2 x = 0$. $\lambda^2 + 2n\lambda + k^2 = 0$.

1°. $n=0$. $x(t) = C_1 \sin(kt) + C_2 \cos(kt) = A \sin(kt + \varphi)$

2°. $0 < n < k$. $\lambda = -n \pm \sqrt{k^2 - n^2} i$ $x(t) = A e^{-nt} \sin(\omega t + \varphi)$, $\omega = \sqrt{k^2 - n^2}$

3°. $n > k$. $\lambda = -n \pm \sqrt{n^2 - k^2}$ $x(t) = e^{-nt} (C_1 e^{\sqrt{n^2 - k^2} t} + C_2 e^{-\sqrt{n^2 - k^2} t})$

4°. $n=k$. $\lambda = -n$. $x(t) = e^{-nt} (C_1 + C_2 t)$



② 非齐次: $h \neq 0$ $n=0$.

1°. $p \neq k$. $x(t) = A \sin(kt + \varphi) + \frac{h}{k^2 - p^2} \sin(pt)$

2°. $p=k$, $x(t) = A \sin(kt + \varphi) - \frac{h}{2k} t \cos(kt)$

共振.

无界!

解法

§ 7.9. 7.10 ODE 其它例题. (Euler 方程. 线性方程组)

Euler 方程

例. $x^3 y''' + x^2 y'' - 4xy' = 3x^2$. 变量. $y^{(n)}$. $a_n x^n$

$$x^3 \frac{d^3 y}{dx^3} + x^2 \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} = 3x^2$$

$$dx^3 = (dx)^3$$

$$x = e^t$$

$$\frac{x}{dx} = \frac{x}{dx} = \frac{1}{\frac{1}{x} dx} \rightarrow d(\ln|x|) = t$$

变为:

$$D(D-1)(D-2)y + D(D-1)y - 4Dy = 3e^{2t}$$

$$\text{即 } D^3 y - 2D^2 y - 3Dy = 3e^{2t}$$

★ 做变换 $x = e^t$, $t = \ln|x|$

Try: $\frac{d^3 y}{dt^3} + \frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} = 3x^2$ X

高阶微分 没有利求不变性
(- 即微分有)

正确做法: $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt} \frac{1}{x}$

思考: x 的阶推? ←

$$\begin{cases} \frac{d^2 y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \\ \frac{d^3 y}{dx^3} = \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right) \end{cases}$$

$$\begin{cases} x \frac{dy}{dx} = \frac{dy}{dt} \\ x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = \frac{d^2 y}{dt^2} \cdot \frac{dt}{dx} \end{cases}$$

Tip: $x^k y^{(k)} = \underbrace{D(D-1) \cdots (D-k+1)}_{k \text{ 个 } D} y$, $D = \frac{d}{dt}$

1° 另解

$$y''' - 2y'' - 3y' = 0 \quad \text{猜 } y = e^{\lambda x}$$

$$x^3 y''' + x^2 y'' - 4xy' = 0 \quad \text{猜 } y = x^\alpha, \quad y' = \alpha x^{\alpha-1}, \quad y'' = \alpha(\alpha-1)x^{\alpha-2}$$

$$y''' = \alpha(\alpha-1)(\alpha-2)x^{\alpha-3}$$

$$x^\alpha (\alpha(\alpha-1)(\alpha-2) + \alpha(\alpha-1) - 4\alpha) = 0$$

$$\alpha(\alpha-1)(\alpha-1) - 4\alpha = \alpha(\alpha^2 - 2\alpha + 1 - 4) = \alpha(\alpha^2 - 2\alpha - 3) = \alpha(\alpha-3)(\alpha+1) = 0$$

$$\Rightarrow \alpha = 0, 3, -1, \text{通解. } y(x) = C_1 x^0 + C_2 x^3 + C_3 x^{-1}$$

2° $x^3 y''' + x^2 y'' - 4xy' = 3x^2$

取一个多项式特解. $y^*(x) = a x^\beta, \beta \text{ 取 } 2. a = -\frac{1}{2}.$

$$\left. \begin{array}{l} C_1 + \frac{C_3}{x} + C_2 x^3 - \frac{1}{2} x^2 \end{array} \right\}$$

例: 3. $x^3 y''' + 3x^2 y'' - 2xy' + 2y = 0.$

$$y = x^\alpha \rightarrow \alpha(\alpha-1)(\alpha-2) + 3\alpha(\alpha-1) - 2\alpha + 2 = 0, \quad \alpha = 1, 1, -2$$

$$= (\alpha-1)^2 (\alpha+2)$$

$$y(x) = C_1 x^{\alpha_1} + C_2 x^{\alpha_2} + C_3 x^{\alpha_3} = \tilde{C}_1 \frac{x^{\alpha_1} - x^{\alpha_2}}{\alpha_1 - \alpha_2} + \tilde{C}_2 x^{\alpha_2} + C_3 x^{\alpha_3}$$

$$\hookrightarrow \tilde{C}_1 x \ln|x| + \tilde{C}_2 x + \frac{C_3}{x^2}$$

$$\left| \begin{array}{l} \frac{d}{d\alpha}(x^\alpha) \\ x^\alpha \ln|x| \end{array} \right.$$

例: 4. $x^2 y'' - 2xy' + 2y = \ln^2 x - 2 \ln x$

$P(\ln x)$

$\hookrightarrow y^*(x) = R(\ln x)$ 代入方程得 R 的一个方程. (R 取多项式), 求出 R . 待定系数法, 特解

3. 例题

例1:
$$\begin{cases} \frac{dy}{dx} = 3y - 2z \\ \frac{dz}{dx} = 2y - z \end{cases}$$

$$y = \frac{1}{2} \left(\frac{dz}{dx} + z \right)$$

$$\frac{dy}{dx} = \frac{3}{2} \left(\frac{dz}{dx} + z \right)$$

$$\frac{dz}{dx} - 2 \frac{dz}{dx} + z = 0$$

$$z = (c_1 + c_2 x) e^x$$

$$y = \frac{1}{2} (2c_1 + c_2 + 2c_2 x) e^x$$

思路: 两个方程, 解方程组?

例2:
$$\begin{cases} x'' + y' - x = e^x \\ y'' + x' + y = 0 \end{cases}$$

3. 例1, "0" 第 5. $D = \frac{d}{dx}$

$$\begin{cases} (D^2 - 1)x + Dy = e^x \\ Dx + (D^2 + 1)y = 0 \end{cases}$$

$$Dx + (D^2 + 1)y = 0$$

$$(-D^4 + D^2 + 1)y = e^x$$

↓

3. 例2

$$\begin{cases} x' = p \\ y' = q \\ p' = \dots \\ q' = \dots \end{cases}$$

$$\vec{r}' = A\vec{r} + \vec{f}$$

$$\begin{pmatrix} y \\ z \end{pmatrix}' = \underbrace{\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}}_A \begin{pmatrix} y \\ z \end{pmatrix}, \quad \vec{r} = \begin{pmatrix} y \\ z \end{pmatrix}, \quad \vec{r}' = A\vec{r}$$

★ e^{Ax} 是什么?

$$e^{Ax} = I + Ax + \frac{A^2 x^2}{2!} + \frac{A^3 x^3}{3!} + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = P \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} P^{-1}$$

$$e^{Ax} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \exp \left[\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} x \right] \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$$

$$\begin{pmatrix} e^x & 4xe^x \\ 0 & e^x \end{pmatrix}$$

$$\begin{pmatrix} e^x & e^x(4x+1) \\ e^x & e^x(4x-1) \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1}$$

$$y(x) = c_1 e^x + c_2 e^x (4x+1)$$

$$z(x) = c_1 e^x + c_2 e^x (4x-1)$$

$$\vec{r}(x) = e^{Ax} \vec{c}$$

$$\begin{aligned} A &= P \Lambda P^{-1} \\ A^n &= P \Lambda^n P^{-1} \\ e^{Ax} &= P e^{\Lambda x} P^{-1} \end{aligned}$$

$$AP = P\Lambda$$

$$P = [P_1, P_2]$$

$$A[P_1, P_2] = [P_1 \Lambda, P_2 \Lambda]$$

$$A P = \lambda P$$

$$y_0 = 1, z_0 = 0$$

$$\begin{aligned} y(0) &= c_1 + c_2 = 1 \\ z(0) &= c_1 - c_2 = 0 \end{aligned} \Rightarrow c_1 = c_2 = \frac{1}{2}$$

$$y(x) = e^x (2x+1)$$

$$z(x) = e^x \cdot 2x$$

$\begin{cases} y(t) = \dots \\ x(t) = \dots \end{cases}$
 page 358.

$$\vec{r}'(t) = A \vec{r}(t) + \vec{f}(t)$$

$$\vec{r}(t) = e^{At} \vec{c}(t)$$

$$= e^{At} \left(\vec{c}_0 + \int e^{-At} \vec{f}(t) dt \right)$$

ODE 总结.

① 基本概念. 所. 解. 通解. "n", 适定性

② 哪些 ODE 可以求出通解?

- $\hookrightarrow \left\{ \begin{array}{l} 1^\circ. \text{ 可分离变量型 ODE} \\ 2^\circ. \text{ 齐次方程} \\ 3^\circ. \text{ 线性方程 (一阶)} \\ 4^\circ. \text{ 高阶方程} \end{array} \right\}$
- 特殊型. 线性. 常系数. 一般. $\checkmark$

③ 有哪些技巧.

$\hookrightarrow$ 变量替换
 $\hookrightarrow$ "猜"

④ 高观点

方程
 $\hookrightarrow$ 解

存在?
 唯一?
 求解?
 (结构)?